RECENT THEORETICAL WORK IN THE UNITED STATES ON FAST REACTOR PHYSICS

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1. Resonance Integral and Doppler Effect Calculations

Since the discovery by Codd and Collins[1] of the importance of fertile–fissile overlap effect in Doppler effect calculations, there has been considerable study of this question in the United States. Hwang,[2] using assumptions similar to those of Rowlands,[1] has given the following approximate expression for the effective cross-section $\tilde{\sigma}_x^{(1)}$ for reaction x of fissile isotope (1) in a homogeneous mixture with fertile isotope (2).

$$\tilde{\sigma}_x^{(1)} = \frac{\left(\tilde{\sigma}^{(e)\,(2)}_{x_{\text{isolated}}} + \tilde{\sigma}_x^{(1)}\right)\left(1 - \dfrac{\langle \Gamma^{(2)} J^{(2)} \rangle}{S^{(2)}}\right)}{1 - \sum\limits_{\text{spin}} \dfrac{\langle \Gamma^{(1)} J^{*(1)} \rangle}{\langle S^{(1)} \rangle} - \dfrac{\langle \Gamma^{(2)} J^{*(2)} \rangle}{\langle S^{(2)} \rangle}}. \tag{1}$$

The notation here follows that of Hwang.[2] The second factor of the numerator represents the overlap effect of isotope (2) on the resonance integral of isotope (1), while the denominator represents a flux correction factor which takes account of the effect of resonances on the magnitude of the group flux. The asterisks on the J^* functions in the denominator of (1) indicate that overlap of other isotopes is included. However, to the order of approximation used here the asterisks can be omitted. For the fertile isotope (2) Hwang writes

$$\tilde{\sigma}_x^{(2)} = \frac{\tilde{\sigma}^{(e)\,(2)}_{x_{\text{isolated}}}}{1 - \sum\limits_{\text{spin}} \dfrac{\langle \Gamma^{(1)} J^{*(1)} \rangle}{\langle S^{(1)} \rangle} - \dfrac{\langle \Gamma^{(2)} J^{(2)} \rangle}{\langle S^{(2)} \rangle}}, \tag{2}$$

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the self-overlap $\tilde{O}_x$ term being negligible in this case, while the fertile–fissile overlap term in the numerator has been dropped because it is not much larger than higher order terms which have also been dropped.

Nicholson,[3] based on work of E. A. Fischer, writes instead for the denominator of (1) and (2)

$$\left(1-\sum_{\text{spin}} \frac{\langle \Gamma^{(1)} J^{(1)} \rangle}{\langle S^{(1)} \rangle}\right)\left(1-\frac{\langle \Gamma^{(2)} J^{(2)} \rangle}{\langle S^{(2)} \rangle}\right)$$

and retains a factor

$$1-\frac{\langle \Gamma^{(1)} J^{(1)} \rangle}{\langle S^{(1)} \rangle}$$

in the numerator of (2). The result of this is that (1) and (2) become

$$\tilde{\sigma}_x^{(1)} = \frac{\sigma_{x_{\text{isolated}}}^{(e)(1)} + \tilde{O}_x^{(1)}}{1-\sum\limits_{\text{spin}} \left(\dfrac{\langle \Gamma^{(1)} J^{(1)} \rangle}{\langle S^{(1)} \rangle} + \dfrac{\tilde{O}_t^{(1)}}{\sigma_p}\right)} \tag{3}$$

$$\tilde{\sigma}_x^{(2)} = \frac{\sigma_x^{(e)(2)}}{1-\dfrac{\langle \Gamma^{(2)} J^{(2)} \rangle}{\langle S^{(2)} \rangle}} \tag{4}$$

where $\tilde{O}_t^{(1)}$ is the overlap term for the total resonance cross-sections.

Thus calculations for each isotope may be carried out ignoring the presence of the other one. The resonance overlap term may be omitted provided flux correction factors are used which are different for each isotope. A similar conclusion has been reached by Hutchins,[4] based on numerical resonance integral calculations for overlapping resonances, except that he omits the denominator in (3), which has only a small effect on the Doppler effect.

Hwang[5] has found that the Nicholson and Hutchins approximations give results for total Doppler effect which agree well with those of formulae (1) and (2) for dilute power reactors, although there may be an appreciable difference for reactors of higher enrichment. The question of whether or not flux correction factors are needed if one chooses to take care of the overlap effect by using the factors in the numerator of (1) and (2) has been discussed in refs. (6) and (7). It was argued in ref. (6) that the errors would not be large. There is, of course, no reason not to use flux correction factors, however. In discussing this subject one must be careful to distinguish between cases in which fertile and fissile material are homogeneously mixed as was assumed above, and in which they are separated, as is sometimes the case in critical assemblies. At energies below about 1 keV, Hwang[8] finds that a statistical treatment is quite unsatisfactory in any case because the level density is not sufficiently large. He has also investigated the accuracy of the narrow resonance approximation, for the homogeneous case, using for comparison an analytical calculation based on a first order approximation to Corngold's slowing down equation,[9] and also a numerical integration of the Boltzmann equation. For the latter purpose he used the MISH MASH code, which is a homogeneous version of RIFF RAFF.[10] The latter provides for cylindrical cell geometry with an isotropic return cell boundary condition, and allows the use of several overlapping isotopes. The conclusion from Hwang's study is that the narrow resonance approximation is in error by about 10% in $\delta\sigma_\gamma$ and $\delta\sigma_f$ and by several per cent in σ_γ and σ_f. The error in the Doppler effect is negligible, however.

The question of effect of level statistics on the Doppler effect has also been pursued by Kelber and Kier,[11] using MISH MASH and a scheme for selecting a sequence of level spacings and resonance parameters. It was found that the dispersion in the fissile material Doppler effect was of the same order as the Doppler effect itself, no matter how many successive sequences were combined. The fissile material Doppler effect itself is quite small, however.

There is general agreement that heterogeneity effects in fast power reactor resonance integral calculations are not large and that the approximate means that have been developed for dealing with them are probably adequate. For critical assembly interpretation, more elaborate techniques are required. The errors in homogeneous resonance integral calculations are not large,[8] but are big enough to make the availability of a direct energy integration of the Boltzmann equation in the resolved resonance region a desirable option in a cross-section preparation routine.

It appears that the greatest potential source of error in fast reactor fertile material Doppler effect calculations is in the computation of low energy flux resulting from inaccurate calculations of elastic removal cross-sections for light elements. Presumably this error can be avoided by use of calculations of the ELMOE type.[12] It seems unlikely that errors due to resonance parameters or resonance integral computation technique would amount to more than about 10%. Comparison between theory and experiment indicates that the theory is underpredicting the Doppler effect by 28% in one case[13] and perhaps even more in another[14] although calculations in the second case have not been completed and it may be that about the same discrepancy will be found in that case also.[15] Underprediction of the low energy flux is the most obvious source of this disagreement, although as will be seen later, it is not obvious where a large increase in calculated low energy flux is to come from.

While the situation for calculation of fertile material Doppler effect is reasonably good, the case is otherwise for the fissile material Doppler effect. Calculations when fertile–fissile overlap is taken into account give a positive effect for Pu-239 equal to about 10% of the negative U-238 one, in a mixture of 7 U-231 to 1 Pu-239, using $\Gamma_f = 0{\cdot}04$ eV for $J = 0$ and $0{\cdot}156$ for $J = 1$,

TABLE 1. PARAMETERS OBTAINED BY FITTING σ_f AND α (widths in eV; $\Gamma_\gamma = 0{\cdot}04$)

$\bar{E}$, eV	$\bar{\sigma}_{f,b}$ barns	$\bar{\alpha}$	$\Gamma n_{j=0}$ (a)	Case 1 $\Gamma_{f_{J=0}} = \Gamma_{f_{J=1}}$, $\nu_f = 2$		Case 2 $\Gamma_{f_{J=0}} = 1{\cdot}5$, $\nu_{f_{J=0}} = 5$, $\nu_{f_{J=1}} = 1$		Case 3 $\Gamma_{f_{J=0}} = 1{\cdot}5$, $\nu_{f_{J=0}} = 10$, $\nu_{f_{J=1}} = 1$		Case 4 $\Gamma_{f_{J=0}} = 0{\cdot}75$, $\nu_{f_{J=0}} = 5$, $\nu_{f_{J=1}} = 1$	
				Γ_f	$\Gamma_{n_{J=0}}$	$\Gamma_{f_{J=1}}$	$\Gamma_{n_{J=0}}$	$\Gamma_{f_{J=1}}$	$\Gamma_{n_{J=0}}$	$\Gamma_{f_{J=1}}$	$\Gamma_{n_{J=0}}$
4300	2·44	0·52	0·0656	0·1208	0·0595	0·0883	0·0539	0·0856	0·0536	0·1012	0·0549
2620	2·80	0·55	0·0512	0·1185	0·0379	0·0906	0·0352	0·0882	0·0351	0·1023	0·0356
1710	3·20	0·58	0·0414	0·1138	0·0272	0·0870	0·0256	0·0848	0·0256	0·0972	0·0259
1200	5·80	0·60	0·0346	0·1063	0·0375	0·0763	0·0347	0·0739	0·0346	0·0866	0·0351
850	6·20	0·62	0·0291	0·1046	0·0271	0·0761	0·0254	0·0739	0·0254	0·0855	0·0257
600	7·00	0·65	0·0245	0·1007	0·0211	0·0746	0·0198	0·0726	0·0198	0·0833	0·0200
420	9·80	0·67	0·0205	0·0970	0·0209	0·0700	0·0197	0·0680	0·0196	0·0785	0·0199
300	13·0	0·70	0·0173	0·0922	0·0201	0·0645	0·0189	0·0628	0·0189	0·0721	0·0191
171	16·0	0·72	0·0131	0·0921	0·0134	0·0669	0·0128	0·0654	0·0127	0·0741	0·0138
80·6	30·0	0·78	0·00898	0·0840	0·0121	0·0575	0·0116	0·0562	0·0116	0·0634	0·0117

(a) Based on *s*-wave strength function of 1×10^{-4}.

and assuming one channel per fission.[13] Experiment indicates a substantially zero effect,[13] so that this uncertainty, while of considerable theoretical interest, should not cause much uncertainty in the total Doppler effect of a dilute power breeder.

Use of new Pu-239 resonance parameters from Harwell,[16] extending up to 300 eV, indicates that the ratio of $\delta\Sigma_\gamma/\delta\Sigma_f$ to Σ_γ/Σ_f, where $\Sigma\gamma$ and Σ_f are macroscopic reactor capture and fission cross-sections, and δ indicates change with temperature, is greater than that given by the unresolved calculation.[13] This would lead to a more negative Doppler effect.

Where there seemed to be some experimental justification for the Γ_f values assumed in ref. (13), the Bohr–Wheeler theory[17, 18] predicts that $J = 0$ resonances should have very broad fission widths not strongly energy-dependent up to a few keV, and the $J = 1$ quite narrow ones with more

TABLE 2. CALCULATED DOPPLER EFFECT AT 1500°K IN UNITS OF 10^{-7} $\delta k/k$

Resonance parameters	U-238/Pu239 ratio	$J = 0$ sequence		$J = 1$ sequence		
		$\partial k_f/\partial T$	$\partial k_\gamma/\partial T$	$\partial k_f/\partial T$	$\partial k_\gamma/\partial T$	$\partial k/\partial T$
Case 1	5	1·80	−0·46	7·10	−1·99	6·45
Case 2	5	1·19	−0·03	4·78	−2·32	3·62
Case 3	5	1·51	−0·02	4·72	−2·33	3·88
Case 4	5	1·91	−0·08	5·05	−2·25	4·63
Case 1	9	2·70	−0·52	10·50	−2·51	10·17
Case 2	9	1·43	−0·03	7·37	−3·01	5·86

energy dependence. This prediction has been confirmed by recent measurements.[19, 20] This would be expected to reduce the Doppler effect over that calculated for the same fission width in both sequences since a considerable part of the fission cross-section would be due to resonances not subject to much Doppler broadening. The DE would arise mainly from narrow resonances. This is the reason for the reduction in Doppler effect noted above for the resolved region, since in that case also most of the fission occurs in very broad resonances. A study has been made[21] in which it was assumed that the ten widest resonances in ref. (28) belonged to the $J = 0$ sequence, and the rest to $J = 1$. The average fission widths for the two sequences based on the available resonance parameters are then 1·52 and 0·059 eV respectively, and chi-squared distribution parameters ν_f 2·98 and 1·1. In the calculations, values of these parameters consistent with those just given were assumed, while others were adjusted to fit the measured σ_f and α of Pu-239. Results of the fitting procedure are given in Table 1, while in Table 2 are given calculated Doppler effects, including self overlap effects. It is seen that a reduction of up to about 50% in Pu-239 Doppler effect was obtained over that with the same Γ_f for both sequences. Even this may not be enough to explain the experimental results, however. Further details may be found in ref. (33).

For U-235 a small positive effect is also calculated.[13] In this case, however, available experiment is in agreement.

2. Analysis of Critical Experiments

A wide variety of pertinent fast neutron critical and ancillary experiments has been performed on the ANL ZPR-III, ZPR-VI, and ZPR-IX facilities. These, coupled with those from hard spectrum Los Alamos Scientific Laboratory experiments, provide a firm base for theoretical correlation and interpretation. However, the experimental data so far are not yet adequate to provide a basis for confirming many important aspects of large fast breeder reactor design. Plutonium (particularly plutonium-240) integral experiments are virtually non-existent, as are also experiments related to fission product effects.

Three comprehensive recent studies[22-24] compared predicted and measured parameters for a wide variety of existing critical assembly data. In general, the reactivity of the known U-235 fuelled systems can be predicted to better than 2% $\Delta k/k$. Only cores having very little core U-238 with significant light element (stainless steel, sodium) core dilution tend to be not as well predicted. In particular, there seems to be an almost linear relationship between the fractional error in predicted reactivity and the atom density of light elements in the core.[23]

Predicted reactivities for both U-235 and Pu-239 fuelled assemblies tend to be somewhat larger than the experimental values. Judicious modifications of cross-sections within stated experimental errors can generally be relied upon to improve agreement between theory and experiment somewhat, but the tendency for overprediction of reactivity, though slight, remains.

There is also considerable inferential data indicating that predicted neutron spectra are more energetic than those measured. The very few proton recoil spectrometer measurements tend to corroborate these findings. Predicted prompt neutron lifetimes tend to be shorter than the measured ones by varying amounts depending upon both cross-section and delayed neutron assumptions.[24] The better predictions seem to be about 10% shorter than experiment.

There is still no satisfactory agreement between theory and experiment on central material replacement data. Such data are usually the preliminary basis for choosing core constituents. Those materials which have some dominant cross-sections (e.g. capture and/or fission) are generally quite reasonably well predicted, but frequently not accurately enough to afford a basis for design. Those which are primarily scattering materials are generally not well predicted, indicating that calculated neutron importance functions are relatively too high at higher energies.[22]

The major problems in correlating theory with experiment are probably due to cross-section uncertainties and improper normalization of data. For example, many fast neutron cross-sections are measured relative to the U-235 fission cross-section. The latter has been lowered by a few per cent due to fairly recent measurements by White *et al.*[25] It is not clear that all analysts have correspondingly altered other reported cross-sections (e.g. capture) to be consistent with the newer absolute measurements of U-235 fission. The full impact of the altered Pu-239 fission cross-sections[25] is not yet apparent, although some tendencies for better correlation between theory and experiment have been noted.[24] Another major source of uncertainty is the 25% deviation in the measurements of U-238 capture cross-section over most of the energy range between 4 and 100 keV.

High energy inelastic cross-sections are also a problem area. Discrete level data are generally available up to ~ 2 MeV. Above this energy, nuclear temperature measurements are the basis for defining inelastic spectra. On the basis of the existing data it is difficult to define inelastic spectra which demonstrate good continuity from level to continuum data.[26] Another problem here is concerned with the fact that the neutron distribution used to obtain the nuclear temperature is at relatively high energies with essentially no data, other than theory, for the distribution of inelastically scattered neutrons in the region between 1 and 50 keV.

The spectrum of neutrons from fission at low energies is another question that has not been extensively studied experimentally. Rather small changes below 50 keV could markedly affect predictions of integral parameters (e.g. neutron lifetime).

It has also been noted[23] that the use of consistent B1 cross-section averaging for light element cross-sections relative to P1 averaging tends to provide a more reactive system. For the bases studied (ZPR-III Assemblies 14, 22, 23, and 32), the reactivity differences range from 0·1 to 1·2% $\Delta k/k$ when comparing the two averagings.

All of the integral correlations between theory and experiment rely heavily upon certain assumptions regarding the experiment. The major ones are cylindrization of the experiment, shape factor calculation, and heterogeneity corrections. As agreement between theory and experiment improves, these rather small corrections will probably be subjected to more intensive study to further facilitate better agreement for confident reactor prediction.

3. Effect of Parameter Uncertainties

Greebler *et al.*,[27] General Electric Co., San Jose, Cal. (GE), have conducted a study of the effect of parameter uncertainties on various quantities of interest for a large pancake core reactor similar to a design concept recently studied by GE. Perturbations in calculated Doppler and sodium void reactivity effects, breeding ratios, fissile enrichment, and neutron spectrum indices were obtained for individual changes in important material cross-sections and composites of such changes. These cross-section changes represented the estimated range of uncertainty as indicated by differences in basic point-wise cross-section data and in standard sets of group constants.[28-31]

Cross-section variations were considered for fuel inelastic scattering, Pu-239 fission, and capture in Pu-239, Pu-240, U-238, stainless steel, and sodium. Lower values of ν for Pu-239 were also considered. Only a limited variation of Pu-239 σ_f was considered, namely, the use of the White data between 24 and 170 keV.[25] The effects produced by these variations on the Doppler and sodium void effects were for the most part minor. A decrease of 16% in the Doppler effect was produced by a high Pu-239 α case based on an analysis by Uttley[32] in which measurements of the Pu-239 total cross-section by Uttley and of the fission cross-section by G. D. James below 20 keV, combined with theoretical evaluation of the Pu-239 scattering cross-sections, give α values between 1·0 and 1·2 over most of the energy region 100 eV to 10 keV. This same variation increased the sodium void effect for total uniform expulsion from the core only by 0·55% k and decreased the breeding ratio by 0·08. This case must be regarded as doubtful, however, until more convincing evidence for such high α values is produced.

Recent evidence that the high Pu-239 α values may not be correct is provided by the critical assembly mockup of the SEFOR reactor in ZPR-III.[33] This was the first all plutonium fuelled assembly having a degraded fast neutron spectrum characteristic of a large dilute fast power reactor. The critical mass of this assembly was found to be only 0·5% higher than that predicted using the base case cross-sections of Greebler *et al.*, except for the lower Pu-239 fission cross-sections of White *et al.* This good agreement may be fortuitous, but it would be spoiled by using the high α values. Also the fact that more rather than less calculated low energy flux seems to be required by Doppler effect measurements and other experiments argues against the validity of the high α values, which were found to cause a 10% reduction in low energy flux.

Variation in σ_γ of U-238 varied the sodium void effect by 0·30% k. Other effects were smaller. A total uncertainty in breeding ratio of about 0·15 appeared possible if the high Pu-239 α case was omitted. A particularly interesting result was that the per cent fissions below 9·1 keV, which is a good measure of the Doppler effect in large reactors, could not be varied by more than $\pm 20\%$ by assuming that all the variations produced a cumulative effect. This indicates that there may be theoretical difficulties in checking experimental results which correspond to a much softer neutron energy spectrum.

A more limited parameter survey was carried out by Hummel *et al.*[34]

of ANL to study the effect of what were considered to be reasonable variations in Pu-239 σ_f, holding α constant, and in U-238 σ_γ on the sodium void effect. The maximum variations produced were 0·4% k for the U-238 σ_γ variation and 1·0% k for the Pu-239 σ_f variation.

The assumed variations in these two studies may not have given a complete picture of the situation for sodium void effect calculations. In the comparison calculations for the 1965 Fast Reactor Conference at Argonne[35] the spread in spectral plus capture components was about 2%, and in particular the ANL results were consistently 1–2% more positive than the GE results. This disparity is not due mainly to differences in Pu-240 or fission product cross-sections, as it also occurred in cases in which those materials were not present. The difference in ANL and GE U-238 capture cross-sections is within the variations studied, and in fact would tend to make the GE results more positive than the ANL ones by up to 0·5% k. Assumptions for Pu-239 α (capture to fission ratio) are similar in the two cases. The only factor identified so far which clearly leads to a more negative GE calculation is that the Pu-239 fission cross-section, which is identical with that used by ANL above 67 keV, assumes considerably higher values in the energy range from 5 to 67 keV, and again below 1 keV. Values similar to the GE ones below 1 keV were used as a variation in the ANL study, and were found to lead to void effects around 0·5% k more negative, because of the lessened energy variation of the adjoint function. Whether the disagreement in fission cross-sections at higher energy could explain the rest of the difference has not been determined at the time of writing. At least part of the discrepancy in results for the spectral component is undoubtedly caused simply by a difference in calculated buckling for a given composition, which would affect the energy variation of the adjoint function.

The uncertainty in the sodium void effect is due largely to uncertainty in the spectral component because of lack of knowledge of the energy variation of fission and capture cross-sections. Calculation of the leakage component is hampered more by the difficulty of performing multi-group, multi-dimensional calculations than by a lack of knowledge of basic data. Critical experiments currently under way at Argonne on ZPR-VI should be very useful in eliminating remaining uncertainties. It seems that the quickest way at the present time to reduce the spectral component uncertainty is to perform central sodium worth measurements in Pu-fuelled central regions in zoned reactors. One such experiment was performed in ZPR-III Assembly 45A, in which the central region mocked-up carbide fuel with a U-238 to Pu-239 ratio of 6. This experiment was analysed by Travelli,[36] who found that the ANL cross-sections used in the comparison calculations gave a very slightly less positive void effect than was measured, an agreement which under the circumstances must be regarded as fortuitous. This also argues against the high Pu-239 α values, however.

Another interesting feature of the comparison calculations was the wide disagreement in the critical mass of very dilute reactors, for which discrepancies in fuel material fission and capture cross-sections become magnified. Uncertainty in the Pu-239 fission cross-section and U-238 capture cross-section is the most likely source of most of the discrepancy. Agreement was much better for less dilute reactors.

4. Reactor Concepts and Performance

Most of the recent conceptual studies in the United States have been related to the neutronics of a 1000 MW(e) fast power breeder reactor system.[37-45] While the emphasis has been on sodium cooled systems, there have been some evaluations and preliminary designs of both steam and gas cooled concepts.[46-48] All of the major integrated design studies utilized the plutonium–uranium-238 cycle. Very little effort has recently been devoted to the uranium-233–thorium cycle in a fast spectrum.

In all the sodium cooled designs except one[38] there was a trade-off between nuclear performance (e.g. breeding) and safety. The balance thus struck has tended to be more conservative on the side of safety in the United States than elsewhere, particularly with respect to the coolant void effect. This has resulted in relatively high ratios of fissile to fertile material in the core, obtained by use of high leakage core geometry. As a result internal conversion ratios have been low, in the neighbourhood of 0·6 to 0·7, and Doppler effects are low unless moderator is added. It is not clear that such compromises will persist in future reactor designs as more is learned about the significance of a positive sodium void effect for the safety of the reactor system. The various design efforts have generated a variety of core configurations utilizing all of the major fuel types that are being developed in the United States. The configurations included annular,[37, 45] pancake,[38, 39, 42] and modular core[40, 44, 45] arrays as well as a conventional core with provision for enhanced radial leakage.[38] A modular core array based on the coupled fast-thermal system has also been recently analysed.[41] The relative merits of the various concepts from a physics and safety point of view have been discussed,[49] but no comprehensive comparison of various core geometries on a consistent basis has yet been made. Some limited comparisons have recently been made,[27, 43, 49] in two cases[27, 43] as sequels to previous studies.[37, 39] In both such cases there was a reaffirmation of the original choice of core geometry, which was a pancake in one case[39] and an annular[37] in the other. Modular designs have been chosen in other studies.[41, 44] It is reasonable to infer from these studies that, at least on the basis of available information, there is not a great deal to choose between pancake, modular, and annular core geometries as a means of limiting a positive coolant effect when all relevant factors are taken into account.

Some specific results from recent studies are as follows. A comparison of pancake and annulus 1000 MW(e) design for nearly the same core diameter gave a sodium void effect which was 0·8% k more positive for total voiding of the pancake core, but only about 0·2% more positive for the worst case.[50] Fissile material inventory requirements for the annular and pancake cores were found in one study to be roughly the same and smaller than that for the modular core.[43] Another study[44] found that with the use of three completely decoupled modules inventory was about the same as for a pancake, while complete voiding of sodium from the core gave an effect 1·5% k more negative.

Economic incentives generally seem to favour design variations which tend to flatten the radial power distribution by such means as the use of multi-enrichment fuels. Both the revised GE pancake[27, 42] and ANL modular design[45] utilize a radial beryllium annulus to obtain some effective flattening. Some designs[45] also use beryllium annuli around the radial blanket to concentrate breeding and simultaneously reduce blanket inventory.

Reactor control is usually accomplished with poison rods. Most designs utilize boron. One recent design utilized tantalum,[42] in order to obtain higher conductivity and better compatibility with stainless steel. Another benefit is the avoiding of the helium gas evolution in boron. Other materials such as rhenium may also be considered. The mixing of moderator with poison has not yet been seriously studied for reactor control.

All of the 1000 MW(e) designs except for the metal fuelled system[45] have an inherent significantly negative Doppler coefficient of reactivity. The metal fuelled system as well as the Westinghouse carbide fuelled system[40] rely on fuel expansion for prompt negative reactivity. The carbide utilized a large, fuelled cermet rod which tends to move one half of the core away from the other.

It is not yet clear to what extent the modular core array has advantages in a coolant loss of flow situation. It is clear, however, that the modular system can tend to have a more significant radial structural expansion coefficient of reactivity; this tends to play a strong role in curtailing adverse effects due to loss of coolant.

Designs are now being evolved to treat the sodium coefficient of reactivity based on the spatial distribution of the coefficient.[51–54] The coefficient is most positive near the centre of the core, becoming negative near the core boundary. The approach taken is to modify the composition of a central zone of the core either by introducing moderator or by altering the isotopic composition of the fuel. Introduction of moderator reduces the energy variation of the neutron importance function and therefore the positive spectral component of the sodium void effect. Use of ZrH_2 moderator in the central 25% of a 4000 litre PuO_2–UO_2 fuelled reactor achieved a reduction in total core sodium void effect of about 1% k at the expense of a reduction of about 0·3 in total breeding ratio.[52] The internal breeding ratio remained relatively high at 0·8, however.

Another approach to spatial design for tailoring the sodium worth is to utilize more than one type of fuel or fuel cycle in a single reactor core. The simplest approach to altering the isotopic composition is to fuel the central region of the reactor with U-235–U-238. This approach would afford a basis for conventional core design which may transfer to a pure Pu–U-238 core when the safety significance of sodium voiding is better understood. Another possibility is to use fresh fuel with a low Pu-240 and fission product concentration in a core zone. Use of this scheme in an annular core design, the inner zone being the inner edge of the annulus, reduced the worst sodium void reactivity change by 0·3% k at the expense of a loss of 0·07 in breeding ratio.[50] The most effective scheme of this type that has been suggested so far is to use U-233–Th fuel in the central region. It was found that for 3000 litre core spherical oxide and carbide fuelled reactors with an 800 litre central zone a reduction in total core sodium void effect of about 2% k was obtained with a 0·10–0·17 loss in breeding ratio[51] for the combined cycles. More recent results on a 4000 litre core oxide fuelled reactor with a 524 litre central zone showed improvement of 1·4% k in the sodium void effect and an actual gain of 0·05 in total combined breeding ratio in a system with internal breeding ratio about 1·0 for both core zones and a maximum positive sodium void effect of about 0·5% k.[53, 54]

The prospect of having a single reactor operating on both the U-233–Th and Pu–U-238 fuel cycle poses some new problems, particularly those associated with the 27- and 3-day transmutation half lives as well as the different core conversion ratios of the two different fuels. The latter can be somewhat adjusted towards equality because of the incentive to provide a radially flat power distribution.

Both the moderated zone and mixed fuel cycle core would lend themselves to provide early conventional design and operation and at the same time provide a reactor design which could be readily altered to a higher performance Pu–U-238 system at a later time. Either concept appears to have advantages with regard to in-core breeding and fuel inventory over the high leakage core geometry, with the disadvantage of added complication, and, in the case of the mixed cycle, of added expense.

A conceptual design study of a 1000 MW(th) helium cooled reactor showed a maximum coolant loss reactivity gain of only 30 cents.[48] It is therefore possible to design such reactors with high internal conversion ratios. Steam cooled concepts, however, have voiding problems similar to those of sodium and parameter uncertainties appear even worse,[35] apparently because of the more degraded spectrum. Only uniform voiding has to be considered, however. The flooding safety of both types of reactors is an important consideration. Calculations have indicated that addition of a small amount of gadolinium to the fuel ensures this for the helium cooled reactor mentioned above.[48] These calculations are very uncertain at present, however. It was concluded from critical experiments[56] that it is feasible to achieve flood safety for small U-235 fuelled steam cooled fast reactors with resonance poisons such as europium and hafnium; recent calculations have confirmed this. Data are needed for softer spectra and Pu fuel, however.

5. Dynamics Studies

Interest in the space-time aspects of the dynamics of fast reactors has been spurred by the trend towards larger reactors and also by a tendency towards the use of weakly coupled cores as a means of combating the coolant void effect problem. There has been a considerable study of coupled modular cores in the U.S. lately.[59–62] While some of this work was concerned with thermal reactor cores, the results are nevertheless of interest for fast reactors. Early studies were those of Avery[63] and of Baldwin.[64] Avery defined for coupled systems 1 and 2 partial neutron densities $N_{11}(t)$ and $N_{12}(t)$, where N_{11} is the density in core 1 of neutrons born in core 1, while N_{12} is the density of neutrons in core 1 that were born in core 2. Then, omitting delayed neutron effects for simplicity of notation,

$$\frac{dN_{11}}{dt} = k_{11}\left(\frac{N_{11}}{l_{11}} + \frac{N_{12}}{l_{12}}\right) - \frac{N_{11}}{l_{11}} \tag{5}$$

$$\frac{dN_{12}}{dt} = k_{12}\left(\frac{N_{21}}{l_{21}} + \frac{N_{22}}{l_{22}}\right) - \frac{N_{12}}{l_{22}} \tag{6}$$

where the k's and l's are multiplication factors and lifetimes. A corresponding set of equations is written for N_{22} and N_{21}. It was pointed out by Gage and Adler[65] in commenting on a similar theory by Hansen[66] that if in (6) one neglects the terms dN_{12}/dt and also neglects the term N_{21}/l_{21} compared to N_{22}/l_{22}, the Avery formulation reduces to that given by Baldwin:[64]

$$\frac{dN_1(t)}{dt} = (k_1 - 1)\,\frac{N_1(t)}{l_1} + \varepsilon N_2(t-\tau) \tag{7}$$

except for the delay time τ occurring in the interaction term $\varepsilon N_2(t-\tau)$ in Baldwin's theory. A similar equation is of course written for N_2. N_1 in this case is the sum of N_{11} and N_{12}, and l_1 is the mean lifetime for the N_1 neutrons. If delayed neutron effects are considered, it is also necessary to neglect their effect in the coupling term to achieve the above reduction.

Analyses of coupled Rover (thermal) reactors[59, 60] indicate that the simpler form of the theory is adequate to reproduce the observed coupling effects, both for oscillator transfer function measurements[60] and for transmission of a pulse from one core to another.[59] The delay time τ seems to be unimportant, certainly for oscillator experiments and even for the pulsed measurements. Whether it could play any role in a rapid fast power reactor transient perhaps remains to be determined.

The coupling in the experiments mentioned above was at most several tenths per cent k, so that it is not surprising that the simpler form of the theory was found to be applicable, as practically all the flux in a core is in the normal mode for that core. Coupling as strong as 4% k was considered[61] for a modular fast reactor.[40] In this study, which used the Avery theory, it was found that the interaction between cores was quite insensitive to what was assumed for the lifetimes l_{12} and l_{21}. This seems to indicate that the simpler form of the theory is adequate in this case also.

The above studies show[58, 61] that a coupling of only a few tenths per cent k is sufficient to produce a significant decrease in the power rise resulting from insertion of reactivity in a given module from what the power rise would be for the same reactivity inserted in an isolated critical module. This is of course not true if the reactivity addition is sufficient to make the individual module become independently critical. The power distribution among modules is quite sensitive to the amount of coupling. No tendency to oscillatory instability has been noted in any of the studies performed so far.

For the dynamics of extended single cores the above techniques probably cannot be used, as the division of all the neutrons in a given region into the category of those born in that region and a second category of those born elsewhere is too much of a simplification.[61] Whether or not the adiabatic approximation[67, 68] has sufficient accuracy for all problems of interest has not been specifically studied for fast reactors. One comparison[69] of the adiabatic approximation for thermal reactors with results of the WIGLE code[70] which provides a numerical space–time integration in two groups indicated satisfactory agreement. This type of comparison is facilitated by using a suggestion of Ergen[71] that the time integral of the space-dependent flux be solved instead of the flux as a function of time. Ergen obtained an analytical solution for this quantity in simple geometries for a one-group problem in which delayed neutron effects are neglected and feedback at any point in space is a function of the integral of the flux at that point, so that heat transfer is neglected. Gyorey and Keck[72] presented a way of altering the usual neutron diffusion equations to obtain a similar solution for the multi-group case. They found good agreement between the WIGLE code and this method, which appears to be very useful for testing the accuracy of the adiabatic approximation.

Measurements[55] of the Avery coupling parameters and prompt lifetime in a coupled fast thermal critical assembly, the mixed spectrum critical assembly,[57] indicated disagreements with theory which are attributed to an underprediction of the low energy neutron spectrum in the fast core. This was confirmed by resonance detector measurements.

An important consideration in mixed oxide fuel development is the degree of fuel homogeneity required for the Doppler effect to be essentially fully

TABLE 3. RATIO OF ENERGY RELEASE FOR SEGREGATED FUEL TO THAT FOR HOMOGENEOUS FUEL

Particle diameter, microns	Period = 0·1 milliseconds				
	E = 0·10, D = 0	E = 0·15, D = 0	E = 0·20, D = 0	E = 0·25, D = 0	E = 0·30, D = 0
10	1·04	1·03	1·03	1·02	1·02
20		1·13		1·09	
30	1·32	1·27	1·23	1·20	1·18
40		1·48		1·35	
60	2·01	1·91	1·81	1·72	1·63
80		2·42		2·15	
100		2·95		2·61	
		E = 0·15, D = 0·15		E = 0·25, D = 0·15	
10		1·07		1·04	
30		1·52		1·34	
60		2·70		2·18	
	Period = 1·0 milliseconds				
		E = 0·15, D = 0		E = 0·25, D = 0	
10		1·00		1·00	
30		1·03		1·02	
60		1·11		1·08	
				E = 0·25, D = 0·15	
10				1·00	
30				1·04	
60				1·14	

E = enrichment.
D = ratio of Doppler coefficient for Pu to that for U-238.

exerted in a transient. This problem has been studied by Fraude[73] and Peterson.[74] Additional calculations were carried out to supplement the work of these authors.[75] As in previous papers, the model assumed was a sphere of Pu-239 enclosed by a U-238 shell. All heat was assumed generated in the Pu-239 and heat loss from the outer shell of U-238 ignored, a reasonable assumption for the 0·1 msec transients assumed pertinent for hypothetical meltdown accidents for fast reactors. Assuming an exponential power rise and neglecting thermal radiation and direct fission product transfer, an exact solution for the temperature can be obtained. The ratio of the energy release for the segregated fuel to that for a homogeneous mixture was then calculated as a function of reactor period, plutonium particle size, plutonium Doppler coefficient, and enrichment. Energy release was calculated to the peak of the burst, which was taken as the time at which the Doppler controlled reactivity equalled the input reactivity. Results are given in Table 3. These studies indicate that fuel segregation is not a significant problem for particles of less than 20–30 μ in diameter in good thermal contact. For processes that start with separate PuO_2 and UO_2 powders it appears that mechanical blending and sintering produce a material which is quite homogeneous on a few micron scale, so that no further homogenization for safety reasons appears necessary. The question of thermal contact and irradiation effects still needs to be investigated.

Acknowledgement

Discussions with Dr. R. N. Hwang have been very helpful in the preparation of section 1.

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Supplement

An investigation has been made[76] of the reasons for discrepancies in sodium void effect and critical size that occurred in the comparison calculations for the October 1965 meeting at Argonne. While not all discrepancies have been satisfactorily explained, it does seem that, as surmised in the main part of the present paper, a major factor was the discrepancy in the low energy Pu-239 fission cross-section. Discrepancy in low energy U-238 capture cross-section was also important in some cases. Discrepancies in assumed Pu-240 and fission product capture cross-sections played a smaller but in some cases appreciable role. Most of the disagreement between the ANL sodium void effect, among the most positive, and that calculated by Atomics International, among the least positive, was explained by variation in the cross-sections mentioned above. Most of the discrepancy between the ANL and GE results could be explained by the higher low energy Pu-239 fission cross-section assumed by GE and by the lower elastic removal cross-sections for sodium apparently assumed by GE.

Smaller discrepancies in sodium void between the ANL results and those of other organizations have not been satisfactorily explained. In some cases the choice of inelastic scattering cross-sections of sodium must have been partly responsible, as the inelastic scattering component as calculated by ANL was considerably higher than that calculated by the other organizations in such cases. This was particularly true of the French, who used the Russian Bondarenko[29] cross-sections. The effect of sodium inelastic scattering cross-section variations has not been explicitly studied so far.

A study made at Atomics International[77] has also confirmed the importance of the choice of low energy Pu-239 fission cross-section in explaining discrepancies in sodium void effect and also in steam void effect for various cross-section sets.

Yet another study of the sensitivity of fast reactor characteristics to cross-section uncertainties made at GE[78] has also indicated the importance of the low energy Pu-239 fission and capture and U-238 capture sections for calculation of the coolant void effect. Significant uncertainties appeared possible also because of uncertainties in capture cross-sections of Pu-240, Pu-241, and fission products, and in Pu-241 fission products. Both sodium and steam cooled systems were investigated. The same trends in coolant void effect were evident in both cases, with larger deviations occurring for a given cross-section variation for steam coolant because of the larger low energy flux.

Supplementary References

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